AHMAD AZEEM

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EDUCATION

Master of Science, Structural and Material Engineering
National Yang Ming Chiao Tung University, Taiwan | Aug 2025 – Expected Jun 2027
Average score (1st & 2nd sem.) - 4.05/4.3 (Received Outstanding Student Award)

Bachelor of Engineering, Civil Engineering
Aligarh Muslim University, India | Aug 2021 – Jun 2025
Percentage: 86.61% (Under top 10 rank in a class of 48 students)

RESEARCH INTERESTS

Computational Geomechanics | Machine Learning | Soil-Structure Interaction | Construction material (Pavement)

RESEARCH EXPERIENCE

- **Dynamic Multi-Objective Optimization of Seismic Port Infrastructure via FEM & AI** ***(July 2026–Present)***

  Research Project | Advisor: Chair Professor E. Pan & C.P Lin, National Yang Ming Chiao Tung University.

  • Developing an automated finite element simulation pipeline in ANSYS to construct a synthetic dataset evaluating non-linear dynamic SSI and seismic permanent displacements of retaining quay walls under multi-parameter earthquake loading.

- **Dynamic SSI of Flexible Foundations in Poroelastic TI Layered Soil** ***(Feb 2026–Present)***

  Research Project | Advisor: Chair Professor Ernian Pan, National Yang Ming Chiao Tung University.

  • Developing a semi-analytical computational framework based on Biot's poroelastic theory and the Stiffness Matrix Method (SMM) to evaluate elastodynamic Green's functions for dynamic soil–structure interaction (SSI).

- **Physics Informed Neural Network for Elastodynamics Green's Function** ***(Sep 2025–Present)***

  Research Project | Advisor: Chair Professor Ernian Pan, National Yang Ming Chiao Tung University.

  • Developing and implementing a Physics-Informed Neural Network (PINN) framework to evaluate elastodynamic Green's functions, providing efficient computational solutions for dynamic Soil-Structure Interaction problem.

- **Advanced Neural Operator and Inversion Modeling for Multilayered Media** ***(Sep 2025–Present)***

  Research Project | Advisor: Chair Professor Ernian Pan, National Yang Ming Chiao Tung University.

  • Developing advanced deep learning architectures, specifically complex-valued Residual Networks and Fourier Neural Operators (FNOs), to resolve elastodynamic behaviors and subsurface property inversion in transversely isotropic half-space multilayered media.

- **Physics-Guided Machine Learning for Pile Uplift Capacity Prediction** ***(Jan 2026– July 2026)***

  Research Project | Advisor: Chair Professor Ernian Pan, National Yang Ming Chiao Tung University. ***(Credentials)***

  • Designing a physics-guided machine learning framework utilizing multilayered, model-agnostic techniques to interpret underlying mechanical behaviors of soil pile interactions and accurately predict the uplift capacity of piles foundation.

- **Explainable AI for strength and mechanistic Interpretation of stabilized soil** ***(Jan 2026–June 2026)***

  Research Project | Advisor: Chair Professor Ernian Pan, National Yang Ming Chiao Tung University. ***(Credentials)***

  • Developed multiple machine learning models and multilayered explainability techniques to explore the mechanistic properties of lime- and cement-stabilized soil and accurately predict its strength.

- **Soil Stabilization and Rockfall Hazard Prediction** ***(Jul 2025 – August2025)***

  Internship | Advisor: Dr. Riya Bhowmik, Indian Institute of Technology, J&K, India ***(Credentials)***

  • Conducted comprehensive experimental soil testing to evaluate the efficacy of various additives in enhancing soil stability and mechanical properties.

  • Predicted rockfall parameters by integrating RocFall 2.0 simulations with AI models.

- **Utilizing the borosilicate glass (BG), PVA and SCMs in Concrete** ***(Aug 2024 – June2025)***

  Bachelor Thesis | Advisor: Prof. Rehan & Prof. Hasan, Aligarh Muslim University. ***(Credentials) (Certificate)***

  • We are using BG powder (10-20%), silica fume (5-15%) and PVA fiber in different percentage in concrete.

  • Part of the work was presented at the ASCE conference.

- **Advancing Green Construction through Alkali Treated Abaca Fibers and Zeolite Integeration** ***(Jul 2024 – Feb 2025)***

Research Project | *Advisor: Prof. Rizwan Ahmad, Aligarh Muslim University.* ***(Credentials)***

- Zeolite was used as a partial replacement for cement and Abaca fiber (0.25-1.5%) was used as reinforcement.
- Found maximum strength at 7.5% zeolite and 0.5% abaca fiber.

▪ **Insights into Road Safety: Literature Review and Comprehensive Data Analysis Approach** ***(Dec 2024 – Jan 2025)***

Internship | *Advisor: Mohan Rao, Council of Scientific and Industrial Research-CRRI,India* ***(Credentials)***

- Analyzed accident data to identify patterns and develop ML models for accident severity prediction.

▪ **Black-Box to Explainable AI: Predicting Compressive Strength of UHPC** ***(May 2024 – Oct 2024)***

Research Project | *Advisor: Dr. Syed Muhammad Ibrahim, Aligarh Muslim University.* ***(Credentials)***

- Developed ML models (DT, RF, GB, AB) to predict Ultra High Performance Concrete (UHPC) compressive strength.
- Applied SHAP plots for feature analysis and model interpretability.
- Provided insights into ML as a viable alternative to traditional UHPC testing methods.

▪ **Leveraging AI to Unlock the Potential of Sustainable Fiber-Reinforced Concrete Materials** ***(May 2024 – Nov 2024)***

Research Project | *Advisor: Dr. Syed Muhammad Ibrahim, Aligarh Muslim University.* ***(Credentials)***

- Designed GB, RF, and ANN models with 1,500 experimental instances.
- Conducted feature impact analysis to enhance model reliability.
- Achieved optimal performance with GB and RF, validated through feature impact analysis.

▪ **Comparative Analysis of Machine Learning Models for Predicting Split Tensile Strength** ***(Dec 2023–Apr 2024)***

Research Project | *Advisor: Dr. Syed Muhammad Ibrahim, Aligarh Muslim University.* ***(Credentials)***

- Developed and compared ensemble ML algorithms for SCM-blended lightweight concrete.
- Achieved high predictive accuracy using Gradient Boosting, validated by MSE, RMSE, and R^2.

PUBLICATIONS (*Research Gate*)

- Explainable AI for Strength Prediction and Mechanistic Interpretation of Lime and Cement Stabilized Soils- Journal of Geotechnical & Geological Engineering- ***A Azeem***, *SS Ansari, E Pan, CP lin*. (10.1007/s10706-026-03720-7)
- Advancing Green Construction through Alkali Treated Abaca Fibers and Zeolite Integration – Journal of building Engineering -***A Azeem***, *JN Akhtar, RA Khan.* (https://doi.org/10.1016/j.jobe.2026.115215)
- Comparative Analysis of Conventional and Ensemble Machine Learning Models for Predicting Split Tensile Strength in Stressed SCM-Blended Lightweight Concrete- *SS Ansari,* ***A Azeem****, M Asad, k Zafar, SM Ibrahim (https://doi.org/10.1016/j.matpr.2024.04.081)*
- Black-Box to Explainable AI: Predicting Compressive Strength of Ultra-High Performance Concrete – ASCE International Conference (Accepted, Paper Id 118)
- Leveraging AI to Unlock the Potential of Sustainable Fiber-Reinforced Concrete Materials – ASCE International Conference (Accepted, Paper Id 189)
- Performance Evaluation of Concrete Containing Borosilicate Glass Powder and Silica Fume – ASCE International Conference (Accepted, Paper Id 303)

PROFESSIONAL EXPERIENCE

Graduate Teaching Assistant, National Yang Ming Chiao Tung University, Taiwan ***(Sept 2026-Present)***

- Facilitating lab sessions, supporting coursework and student evaluation for Foundational Engineering (516221).
- Supporting coursework and student evaluations for Scientific Paper Writing (536224).

Seed Teaching Assistant, TSIR-National Institute of Applied Research, Taiwan ***(Jul 2026-Present)***

- Assisting with educational camps, workshops, and science outreach programs, as well as facilitating global exchanges.

Graduate Research Assistant, National Yang Ming Chiao Tung University, Taiwan ***(Sept 2025-Present)***

- Developing advanced machine learning models and semi-analytical approaches to analyze soil-structure interactions.

Undergraduate Research Assistant, Aligarh Muslim University, India ***(Aug 2024-June 2025)***

- Executed comprehensive experimental testing to support the development and structural evaluation of sustainable concrete.

Site Engineer, Dwarika Infocom Pvt. Ltd., Noida, India ***(Jan 2021–June 2021)***

- Supervised field operations ensuring adherence to design, quality, and safety standards.

Intern, John Holland, Australia ***(Feb 2024–Mar 2024)***

- Calculated concrete volumes for bridge pile caps and abutments; compared subcontractor rates.
- Evaluated building drawings to streamline task sequencing and improve project efficiency.

SKILLS

- **Software:** STAAD Pro, AutoCAD, Revit, Matlab, Ansys, ELPLA
- **Programming language:** Python, R, Html
- **Material Testing:** UPV test, Electrical resistivity, SEM analysis, Triaxial shear, Proctor compaction, Direct shear test

AWARDS AND ACHIEVEMENTS

(Credentials)

- **2026:** Excellent international volunteer awards for serving on NYCU International day**.**
- **2026**: Honor student distinction award for outstanding performance in 1st semester of MS.
- **2026-2027:** Outstanding existing student scholarship award for maintaining outstanding grades by NYCU.
- **2024:** Merit-cum-Means scholarship by Bharat Sewa Trust.
- **2021-2025:** Merit-cum-Means scholarship for Bachelor of Engineering by State government.
- **2017 – 2020:** Merit-cum-Means scholarships (3 consecutive years) by Central government.
- **2014:** 1st Prize, Quiz Competition sponsored by 3S classes.
- **2013:** 2nd Prize, Essay Writing Competition.

WORKSHOPS & TRAINING

(Credentials)

- Presented three research paper at the ASCE International Conference — *PEC, India (Feb 2025)*
- CFD simulation in Civil and Environmental Engineering — *ACI, America (Mar 7,2024)*
- Workshop on Disaster Risk Reduction in Civil Engineering Structures — *AMU, India (Oct 2022)*
- Short-term Training on Disaster Management & Mitigation — *NIDM & ZHCET, AMU (Nov 2022)*

VOLUNTEER WORK

(Credentials)

- Served as a event planning team member of International Student Association, NYCU for fall semester.
- Served as a National Cadet Corps (NCC) member for three consecutive years.
- Contributed as a Coordinator and volunteer in the NGO, Wing of Desire for 1 year.
- Contributed as a volunteer in the NGO, Aligarh Renaissance for 1 year.

CO-CURRICULAR ACTIVITIES

(Credentials)

- Selected as a speaker from the entire civil engineering department at the International Student and Faculty Gathering events.
- Served as a MC for the International Student Gathering, organized by the College of Engineering, NYCU, in June 2026.
- Led a group of 10 students as the group leader in a survey camp, in Amu, India
- Participated in the 'Eco Fox Run' to promote a clean and green environment, conducted by the Eco Club Amu, India.
- Qualified for the National Muslim Talent Search Examination.

MOOCs

(Credentials)

- EMI teaching assistant training course. — *(Fullbright Taiwan, Sept, 2026)*
- Finite element modelling in Matlab — *(MathWorks, Nov, 2025)*
- Artificial Intelligence applications for Civil Engineers. — *(Udemy, Nov,2024)*
- 100 days of Code: The complete Python Pro bootcamp. — *(Udemy, Apr, 2024)*

RECOMMENDATION

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